AQ10.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

Consider a function $f: R^2 \to R$ defined as $f(x, y) = x^2 + 2xy^2 + 2x^2y + y^2$. If $\nabla f(1, 1) = (a, b)$, then find the value of $a - b$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Consider the following function defined as

$$f_1(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 1 & \text{if } y = 0. \end{cases}$$

$$f_2(x, y) = xy + x^2y^2$$

- **Statement 1:** f_1 has continuous gradient at the origin but f_2 does not have continuous gradient at the origin.
- **Statement 2:** f_2 has continuous gradient at the origin but f_1 does not have continuous gradient at the origin.
- **Statement 3:** $f_1(x, y)$ and $f_2(x, y)$ have continuous gradients at the origin.

Which of the above statements is correct?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point

Consider a function $f: R^2 \to R$ defined as $f(x, y) = e^{(x^2 + xy + y^3)}$.

Which of the following option(s) is (are) true?

- [ ] $\nabla f = (e^{(x^2 + xy + y^3)}, e^{(x^2 + xy + y^3)})$
- [ ] $\nabla f = (2x + y, 3y^2 + x)$
- [ ] $\nabla f = (2xe^{(x^2 + xy + y^3)}, 3y^2 e^{(x^2 + xy + y^3)})$
- [ ] $f(x, y)$ increases most rapidly in the direction of $\frac{1}{5}(3, 4)$ at the point (1,1).
- [ ] $f(x, y)$ decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ at the point (1,-1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y)$ increases most rapidly in the direction of $\frac{1}{5}(3, 4)$ at the point (1,1).
$f(x, y)$ decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ at the point (1,-1).

Level 2:

With a particular frame of reference (in R^2), a corner of a hot rectangular iron plate with some finite length and breadth is placed at the origin $(0, 0)$. The temperature of the plate at a point (x, y) is given by the function:

$$T(x, y) = xy^2 + x^2y$$

Use this information to answer the following questions

1 point
From the point (1,3), in which direction the rate of change in temperature is the minimum?

- ○ $\frac{1}{\sqrt{274}}(-15, -7)$
- ○ $\frac{1}{\sqrt{82}}(9, 1)$
- ○ $\frac{1}{\sqrt{82}}(-9, -1)$
- ○ $\frac{1}{\sqrt{274}}(15, 7)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{274}}(-15, -7)$

1 point
From the point (1,3), in which direction the rate of change in temperature is the maximum?

- ○ $\frac{1}{\sqrt{274}}(-15, -7)$

○
$\frac{1}{\sqrt{82}}$ 82
√
1(9, 1)
○
$\frac{1}{\sqrt{82}}$ 82
√
1(-9, -1)
○
$\frac{1}{\sqrt{274}}$ 274
√
1(15, 7)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{274}}$ 274
√
1(15, 7)

1 point

From a point (1,3), in which direction the rate of change in temperature of the iron plate is zero?

☐
$\frac{1}{\sqrt{274}}$ 274
√
1(-7, 15)
☐
$\frac{1}{\sqrt{82}}$ 82
√
1(1, -9)
☐
$\frac{1}{\sqrt{82}}$ 82
√
1(-1, 9)
☐
$\frac{1}{\sqrt{274}}$ 274
√
1(7, -15)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{274}}$ 274
√
1(-7, 15)
$\frac{1}{\sqrt{274}}$ 274
√
1(7, -15)

1 point

Consider a function $f: R^2 \to R$f : R2 → R defined as $f(x, y, z) = 2e^{2x} \sin yz$
f(x, y, z) = 2e2x sin yz. At the point (0, $\frac{\pi}{2}$, 12π, 1), which are the directions that have the directional

derivative to be 0 ?

- [ ] $\frac{1}{\sqrt{2}}(0,1,1)$
- [ ] $\frac{1}{\sqrt{2}}(0,1,-1)$
- [ ] $\frac{1}{\sqrt{3}}(1,1,1)$
- [ ] $\frac{1}{2021\sqrt{2}}(0,2021,-2021)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{1}{\sqrt{2}}(0,1,1)$

$\frac{1}{\sqrt{2}}(0,1,-1)$

$\frac{1}{2021\sqrt{2}}(0,2021,-2021)$

1 point

Consider a function $f: \mathbb{R}^3 \to \mathbb{R}$ defined as $f(x,y,z) = \alpha x + \beta x y^2 + \gamma y z^2$. If the maximum value of the directional derivative of $f(x,y,z)$ at (1, 1, 1) is in the direction parallel to x- axis and has magnitude 5, then

- ○ $f(x,y,z) = 5x + x^2$
- ○ $f(x,y,z) = 2x + xy^2$
- ○ $f(x,y,z) = 5x$
- ○ $f(x,y,z) = xy^2 + 2yz^2$
- ○ $f(x,y,z) = 2x$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f(x,y,z) = 5x$

Check Answers